Lecture 5: What Is It Like to Be a Baby: The Development of Thought (full transcript)

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Overview: This lecture explores issues and ideas related to the branch of psychology known as cognitive development. It begins with an introduction of Piaget who, interested in the emergence of knowledge in general, studied children and the way they learn about the world in order to formulate his theories of cognitive development. This is followed by an introduction to the modern science of infant cognition. Finally, the question of the relationship between and the existence of different kinds of development is addressed.

Chapter 1. Jean Piaget, Stage Theory and Its Limits [00:00:00]

Professor Paul Bloom: So, most of what we do these days – our methods, our theories, our ideas – are shaped, to some extent, by Piaget's influence. And so, what I want to do is begin this class that's going to talk about cognitive development by talking about his ideas. His idea was that children are active thinkers; they're trying to figure out the world. He often described them as little scientists. And incidentally, to know where he's coming from on this, he had a very dramatic and ambitious goal. He didn't start off because he was interested in children. He started off because he was interested in the emergence of knowledge in general. It was a discipline he described as genetic epistemology – the origins of knowledge. But he studied development of the individual child because he was convinced that this development will tell him about the development of knowledge more generally. There's a very snooty phrase that — I don't know if you ever heard it before. It's a great phrase. It's "Ontogeny recapitulates phylogeny." And the idea of this — What that means is that development of an individual mimics or repeats development of the species. Now, it's entirely not true, but it's a beautiful phrase and Piaget was committed to this. He was very interested in saying, "Look. We'll figure how a kid develops and that will tell us about the development of knowledge more generally."

So, Piaget viewed the child as a scientist who developed this understanding, these schemas, these little, miniature theories of the world. And they did this through two sorts of mechanisms: assimilation and accommodation. So, assimilation would be the act of expanding the range of things that you respond to. Piaget's example would be a baby who's used to sucking on a breast might come to suck on a bottle or on a rattle. That's changing the scope of things that you respond to. Accommodation is changing how you do it. A baby will form his mouth differently depending on what he's sucking on. And so, these processes where you take in — I'm giving this in a very physical way, but in a more psychological sense you have a way of looking at the world. You could expand it to encompass new things, assimilation. But you could also change your system of knowledge itself – accommodation. And Piaget argued that these two mechanisms of learning drove the child through different stages. And he had a stage theory, which was quite different from the Freudian stage theory that we have been introduced to. So his methods were to ask children to solve problems and to ask them questions. And his discoveries that — they did them in different ways at different ages led to the emergence of the Stage Theory.

So, for Piaget, the first stage is the sensorimotor stage or the sensorimotor period. For here the child is purely a physical creature. The child has no understanding in any real way of the external world.

There's no understanding of the past, no understanding of the future, no stability, no differentiation. The child just touches and sees, but doesn't yet reason. And it's through this stage that a child gradually comes to acquire object permanence.

Object permanence is the understanding that things exist when you no longer see them. So those of you in front, you're looking at me and I go [ducks behind lectern]. It occurred to me it'd be a great magic trick if I then appeared in back. But no, I'm just here. That's object permanence. If I went under here and then the people said, "Where the hell did he go? Class is over," that would show a lack of object permanence. So, adults have object permanence. Piaget's very interesting claim is that kids don't. Before six-month-olds, Piaget observed, you take an object the kid likes like a rattle, you hide it, you put it behind something, it's like it's gone. And he claimed the child really thinks it's just gone. Things don't continue to exist when I'm not looking at them anymore. And so he noticed they — they're surprised by peek-a-boo. And Piaget's claim was one reason why they're surprised at peek-a-boo is you go — you look at a kid, the kid's smiling and go, "Oh, peek-a-boo," and you close — and you cover your face and the kid says, "He's gone." "Peek-a-boo." "Oh, there he is. He's gone." And you really — That's the claim.

Piaget also discovered that older children fail at a task that's known as the A-not-B task. And Peter Gray in his psychology textbook refers to it as the "changing hiding places" problem, which is probably a better name for it. And here's the idea. You take a nine-month-old and for Piaget a nine-month-old is just starting to make sense of objects and their permanence. You take an object and you put it here in a cup where the kid can't see it, but it's in the cup. So the kid, if you were the kid, will reach for it. You do it again, reach for it. You do it again, reach for it. That's point A. Then you take — you move it over here. Piaget observed kids would still reach for this. It's like they're not smart enough to figure out that it's not there anymore, even if they see it move. And this was more evidence that they just don't understand objects, and that this thing takes a lot of time and learning to develop.

The next stage is the preoperational stage. The child starts off grasping the world only in a physical way, in a sensorimotor way, but when he gets to the preoperational period the capacity to represent the world, to have the world inside your head, comes into being. But it's limited and it's limited in a couple of striking ways. One way in which it's limited is that children are egocentric. Now, egocentrism has a meaning in common English which means to be selfish. Piaget meant it in a more technical way. He claimed that children at this age literally can't understand that others can see the world differently from them. So, one of his demonstrations was the three mountains task. We have three mountains over there. You put a child on one side of the mountains and you ask him to draw it, and a four- or five-year-old can do it easily, but then you ask him to draw it as it would appear from the other side and children find this extraordinarily difficult. They find it very difficult to grasp the world as another person might see it.

Another significant finding Piaget had about this phase of development concerns what's called "conservation." The notion of conservation is that there's ways to transform things such that some aspects of them change but others remain the same. So, for instance, if you take a glass of water and you pour it into another glass that's shallow or tall, it won't change the amount of water you have. If you take a bunch of pennies and you spread them out, you don't get more pennies. But kids, according to Piaget, don't know that and this is one of the real cool demonstrations. Any of you who have access

to a four- or five-year-old, [laughter] a sibling or something — Do not take one without permission, but if you have access to a four- or five-year-old you can do this yourself. This is what it looks like. The first one has no sound. The second one is going to be sound that's going to come on at the end [plays video]. But there's two rows of checkers. She asks the kid which one has more. The kid says they're the same. Then she says — Now she asks him which one has more, that or that. So that's really stupid. And it's an amazing finding kids will do that and it's a robust finding.

Here's another example. So, they're the same [tape playing]. So, it's a cool finding of that stage, suggesting a limitation in how you deal and make sense of the world. The next phase, concrete operations, from seven to twelve, you can solve the conservation problem, but still you're limited to the extent you're capable of abstract reasoning. So the mathematical notions of infinity or logical notions like logical entailment are beyond a child of this age. The child is able to do a lot, but still it's to some extent stuck in the concrete world. And then finally, at around age twelve, you could get abstract and scientific reasoning. And this is the Piagetian theory in very brief form.

Now, Piaget fared a lot better than did Freud or Skinner for several reasons. One reason is these are interesting and falsifiable claims about child development. So claims that — about the failure of conservation in children at different ages could be easily tested and systematically tested, and in fact, there's a lot of support for them. Piaget had a rich theoretical framework, pulling together all sorts of observations in different ways, wrote many, many books and articles and articulated his theory very richly. And most of all, I think, he had some really striking findings. Before Piaget, nobody noticed these conservation findings. Before Piaget, nobody noticed that babies had this problem tracking and understanding objects.

At the same time, however, there are limitations in Piaget's theory. Some of these limitations are theoretical. It's an interesting question as to whether he really explains how a child goes from a concrete thinker to an abstract thinker, or how he goes from not having object permanence to understanding object permanence. There's methodological limitations. Piaget was really big into question and answer, but one problem with this is that children aren't very good with language, and this might lead you to underestimate how much they know. And this is particularly a problem the younger you get.

Methodology is going to loom heavy in the discussion of any science and that includes psychology. Often 90% of the game is discovering a clever method through which to test your hypotheses. We're going to talk a little bit about that regarding babies. I'll give you another example from a very different domain. There was a set of scientists interested in studying tickling. So, when you tickle somebody, under what circumstances will they laugh? Where do you have to tickle them? Can you tickle yourself? Does it have to be a surprise, and so on? It turns out very difficult to study this in a lab. You're not going to have your experimental credit. You come into the lab and say, "Okay. I'm the graduate student. Ha, ha, ha." And [laughter] in fact, an example of a methodological attempt was done by Henry Gleitman at University of Pennsylvania, who built a tickle machine, which was this box with these two giant hands that went "r-r-r-r." This was a failure because people could not go near the tickle machine without convulsing in laughter. But we will discuss when we have a lecture on laughter a bit of the tickle sciences.

And finally there's factual. What do infants and children really know? It's possible that due to the methodological limitations of Piaget, he systematically underestimated what children and babies know. And in fact, I'll present some evidence suggesting that this is in fact — that this is the case.

Chapter 2. The Modern Science of Infant Cognition [00:12:50]

So, I want to introduce you to the modern science of infant cognition. Infant cognition has been something studied for a very long time. And there was a certain view that has had behind it a tremendous philosophical and psychological consensus. And it's summarized in this Onion headline here. And the idea is that babies are stupid, that babies really don't know much about the world. Now, the work that this Onion headline is satirizing is the recent studies, which I'm going to talk about, suggested that on the contrary, babies might be smarter than you think. And to discover the intelligence of babies we have to ourselves be pretty smart in developing different techniques.

To study what a baby knows, you can't ask your questions. Babies can't talk. You could look at what it does but babies are not very coordinated or skilled so you need to use clever methods. One clever method is to look at their brain waves [laughter]. This child on the right died during testing. It was a tragic — It was crushed by the weights [laughter] of the electrodes. He's happy though. You could study their brain waves. One of the few things babies can do is they could suck on a pacifier. And you might think, well, how could you learn anything from that? Well, for instance, you could build machines that when babies suck on a pacifier they hear music or they hear language, and then you could look at how much they suck on the pacifier to determine what they like.

But undeniably we know most of our — we got most of our knowledge about babies from studies of their looking times. That's one thing babies can do. They can look. And I have up here — This is a picture of Elizabeth Spelke, who is a developmental psychologist who's developed the most research on looking at babies' looking times and what you could learn from them. And I have here two ways you could learn from looking. One is preference. So for instance, suppose you want to know, for whatever reason, do babies like the looks of dogs or cats? Well, you could put a baby down, have a picture of a dog here, a picture of a cat here, and see which one the baby looks at. Babies can move their eyes and that could tell you something. Do babies distinguish pretty faces from ugly faces? Well, put a pretty face here, an ugly face here, see if the baby prefers to look at the pretty one. You could also do habituation and surprise. And much of the studies I'm going to talk about here involve habituation and surprise.

Habituation is a fancy word for boredom. What you do is you show a baby something over and over again. Now, remember from behaviorism the baby will learn this isn't very interesting. Then you show the baby something different. If the baby really sees it as different, the baby will look longer, and you could use that as a measure of what babies find different. For instance, suppose you want to know if the baby can tell green from red. Well, you could show the baby a green patch, a green patch, a green patch, a green patch; the baby'll get bored, then a red patch. If they all look the same to the baby, the baby will just continue to tune out, but if the red looks different the baby will perk up. And this is, in fact, one way they study color vision in babies.

Surprise is related to this. You could show babies something that shouldn't happen. If babies are like — If babies also think it shouldn't happen, they might look longer, and essentially what happens is

scientists do magic tricks to explore this very thing. And to start with some real examples, a lot of this infant research has gone back to the Piagetian question of object permanence, asking, "Is it really true babies don't know that objects remain even when they're out of sight?" So one very simple study by Spelke and Baillargeon: Have babies shown a block with a bar going back and forth like that. So the bar just goes back and forth. Now, there's something you do that's so obvious you probably don't even know you're doing it. When you see a display like that, what you assume is there's a bar there, and what that means is there's something in the middle that you've never seen before. But of course, if you were a simple perceptual creature, you would just see that there'd be a bar on top and a bar on the bottom. You wouldn't expect anything in the middle because you never saw anything in the middle. So, what you do then is you show babies this and then you show them either B or C and if we do this with adults you expect B, C is almost a joke. And, in fact, babies respond the same way. Babies expect there to be an entire, complete bar and are surprised and look longer at the broken bar.

Other studies, some of them — Well, here's another study by Rene Baillargeon looking at the same thing in a different way. You show the baby, say a six-month-old, a stage with a block on it. Then a screen rises and obscures the block. Now, if the babies expect the block to still be there, they should think the block should stop the screen. On the other hand, if out of sight out of mind, they should expect the screen to keep going. So, what you do is you set up a couple of displays, one where the block is stopped, the other one where you take this away with a trap door and it keeps going. And, as you see, the baby screams when this happens. That doesn't really happen, but they do look longer.

One final example of an object permanence study. Some of this work's been done at Yale in Karen Wynn's lab, where they look at babies' understanding of addition and subtraction. And a lot of it is done with real objects, but there's also animated versions so here is an animated example [tape playing]. Babies are surprised. They expect 2 - 1 = 1 and when 2 - 1 = 2 or 3 or 0, they look longer, indicating surprise. And even six-month-olds are sensitive to these rudimentary facts of arithmetic, telling us something about their mathematical knowledge, but also telling us something about that they expect things to remain when they're out of sight.

Now, this research suggests that infants' understanding of the physical world is there from the very start, but at the same time not entirely. We know there are certain things babies don't know. Here's an example. Suppose you show babies this. You have a block here and then you have something above there floating in mid air. Babies find this surprising. Even six-month-olds find this surprising. It violates gravity, but six-month-olds aren't smart enough to know that a block just stuck over here is also surprising. Twelve-month-olds will think that it should fall. Six-month-olds don't, and even 12-month-olds don't find anything weird about this, while adults are sophisticated enough to understand that that's an unstable configuration and should fall over. So, although some things are built in, some things develop.

And this raises the question of, "How do we explain development?" How do we explain when babies come to know things that they didn't originally know? Well, one answer is neural maturation, growth of the brain. Most of the neurons you have now in your head, right now, you had when you were in your mother's uterus. What happens in development isn't for the most part the growth of new neurons. It's for the most part pruning, getting rid of neurons. So, the neural structures change radically as babies kind of get rid of excess neurons through development. At the same time though, connections

between neurons grow like crazy and they — and this process of synaptic growth where there are the connections across different synapses peaks at about two years. Finally, remember myelination, where you sort of get this fatty sheath over your neuron to make it more effective? That also happens through development, and in fact, it goes through development and even teenagers are not fully myelinated. In particular, they're not fully myelinated in their frontal lobes. Recall that frontal lobes are involved in things like restraint and willpower. And so, it could be the problem is the baby's brain doesn't develop yet.

Another possibility is there's problems with inhibition. This is related, again, to the frontal lobes and this comes out with the A, not B error. So, remember the baby reaches, reaches, reaches. It's moved, reach, follow, keeps reaching the same place. And it could be that babies don't know anything about objects. But another possibility is once you do something it's kind of hard to stop. It takes a bit of control to stop. And there's all sorts of independent evidence that babies lack this control. The part of their brain that could control certain behaviors is just not active yet.

There's a very nice illustration of inhibitory problems from a "Simpsons" episode that actually sort of covers anything you might want to know about developmental differences. So it goes like this [tape playing]. And that basically may sum up much of developmental psychology. That the child essentially — he does A, A, A. It's moved. You go, "doh!" and he keeps going for it. And there's some evidence that's true. Adele Diamond who studies this finds that although kids reach for A, they look for B, as if they know it's there but they can't stop themselves from reaching. And we'll continue this theme a little bit later.

Finally, it might be kids don't know things. Some things you've got to learn. And this is true in all sorts of domains – in the social world, in the economic world, in the political world – and it's true as well in the physical world. In fact, there's some things even adults don't know. So, here's a study by Michael McCloskey with college students. Here's the idea. You have a tube, a transparent — a tube — a hollow tube, and at the top of the tube you throw a ball through so it whips through the tube and it comes out. The question is, "What happens to it?" Does it go in the path of A, or does it go in the path of B? Without looking around, who votes for A? Who votes for B? Here's the weird thing. Whenever I do this at Yale everybody gets the damn thing right [laughter]. At Johns Hopkins, 50/50, [laughter] for A and B. I got to get a better demo. But anyway, college students not here, show systematic biases of incorrect physical intuitions. Here's a twist, and if you found people who were less wonderful than you all, and asked them you'd get a lot of people saying the curving thing. But here's a twist. Ask somebody, "What if you took a tube and you squirted water through it? Where would the water go?" Nobody chooses B. Everybody knows the water would continue in a straight line, suggesting that when you have experience that helps you out, but in absence of experience you're kind of lost.

Chapter 3. Babies in the Social World [00:25:14]

We've talked about the physical world. What about the social world? What about the world of people? Well, there's a lot of research on this as well. Babies start off with some social preferences. If you take newborn babies — It's very hard to do research with newborn babies actually because of the consent procedure and everything, so most of this work is done in France [laughter], where they have no laws at all. They just rush in to — Women give birth and they rush in and they say, "We are psychologists,"

and then we do experiments on the babies, and it's terrific. And this is one of them where they compare babies looking at this versus this. Babies like the one that looks like a face. These are newborns. There are some preferences with humans and with other primates to favor faces. Babies are also social animals too, so they're natural mimics.

Andrew Meltzoff, for instance, has found that if you go to a newborn baby, and if you find a newborn baby, this is the first thing you should do. Stick your face right up to the newborn baby and go like this and stick your tongue out. And Meltzoff finds that babies more often than not stick their tongues out back, suggesting some sort of social connection from one person to another, and then later on babies are mimics. Babies more often than not will copy the face next to them. Now, these — the nature of these responses, this preferring faces, this sort of mimicry, is a matter of debate, and there's a lot of research going on asking how smart are babies. Can we see — use some of the same methods that we've looked at for the physical world to look at the social world?

And to illustrate one of the studies, I'll tell you about a study that I did with Valerie Kuhlmeier and Karen Wynn. And so, what we tested was nine-month-olds and twelve-month-olds, and we showed them movies. So, they're sitting down and they're seeing a movie where one character's going to help a ball achieve a goal, and another character's going to hinder the ball. And then we're going to see whether they expect the ball to approach the one that helped it versus the one that hindered it. So, this is what a baby would see. This is literally the same movie a baby would see in the experiment. The thing is for these sorts of experiments there is a lot of control, so something that's a square in one movie will be a triangle in another movie; something that's on the top in one movie will be on the bottom in another movie. So, this is an example movie but this is what babies would see. [video playing]

And they'd see this over and over again and the question is would they expect babies — would babies expect the one to approach the one that helped it or approach the one that hindered it? And what we find is, statistically, babies look longer when shown a movie where it approaches the one that hindered it versus helped it. And this we take as preliminary evidence that they have a social interpretation. They see this movie as you see this movie in terms of helping and hindering, and somebody going to somebody that helped it versus hindered it. You could then ask — This makes a prediction that babies should themselves prefer the creature who's the helper versus the hinderer, and to explore this, a graduate student in this department, Kiley Hamlin, has started a series of studies where they show babies three-dimensional scenes and then give them the characters and see which one they reach for. So, here's video so you could see how this experiment is done. [video playing]

Now, the next trial is from a different study. A different thing we use, and the baby is given a choice. One thing to know methodologically is the person giving a choice is blind to the study. And blind here is a technical term meaning she had no idea what the baby saw, and the point about this is to avoid either intentional or unintentional sort of trying to get the answer you want. She couldn't do that because she didn't know what the right answer is. So, here's what the baby would see. [video playing]

So, this suggests that some social understanding may be there from the very start. This evidence is tentative, very controversial.

But now, I want to raise a huge developmental puzzle and the puzzle is there are some ways in which babies are — not just babies, but young children are very clueless when it comes to people. And so, I have a film clip here of two very nice studies showing babies' ignorant — sorry, young children's ignorance of other people. I'll show you the studies and then we'll briefly discuss what they mean. Uh oh. [video playing]

Chapter 4. Question and Answer on Learning and Development [00:31:26]

Professor Paul Bloom: Before discussing that example in a little bit more detail, any questions? What are your questions? Yes, in back.

Student: [inaudible]

Professor Paul Bloom: Typically — I don't know for those particular children, but typically on those tasks three-year-olds and young four-year-olds tend to fail, and around the age of four or five kids tend to succeed. There's sort of a period around the age of four, four and a half, where kids make the transition from failure to success. The question, by the way, was when do children — in that video when were the — what were the ages of the children who failed and who passed? Yes.

Student: [inaudible]

Professor Paul Bloom: The question of whether discriminant conditioning has been used with babies to explore what sort of concepts they have. I don't know. Does anybody — It has been —

Graduate Student: — It's not as effective —

Professor Paul Bloom: Koleen answered and said that it's not as effective as other methods. Part of the problem with using operant conditioning with babies is it's difficult to get them to behave in any systematic way. So, the looking-time measures tend to be more subtle. Any other questions? Oh. Yes.

Student: [inaudible]

Professor Paul Bloom: Oh. The question of why they chose — the baby — the kids chose the rocket ship one as opposed to the Rafael one. It wasn't what they were interested in in the experiment. And my bet is when they chose the stickers they had a pretty good sense of why, of which ones the boys would prefer in those studies. The question of why a boy might prefer one sort of sticker, and you might get a different response with a girl, is going to come up later when we discuss different theories of sex differences. But that was something I think they were just assuming in the study to get it off the ground. Okay.

Chapter 5. Review of Studies Presented in Class; Autism [00:33:54]

There's a huge debate over what's going on there. And if you listened at the end to the psychologist summarizing the data, the psychologist had a very good and very clear and strong idea of what was going on. It was that children need to know more about minds. The children don't know about that you can do something with the intent to deceive. They don't understand that somebody could choose what you chose in a malicious way. This is possible. This is one respectable theory, but the alternative is they have the right knowledge, but they suffer from problems with inhibition.

So, consider both studies. The first study, the one with the deceptive dolls with the big shoes and little shoes, is actually fairly difficult. And it's possible that children kind of got overwhelmed with it, and when asked what would the mother think, who the mother would think stole the food, responded with who really stole the food. And that there's some pull towards the right answer that makes this task difficult.

The second one — the second study illustrates this issue even more clearly. Take the boy who kept failing. He kept pointing to the rocket ship and mean monkey kept taking it away. It's possible that he genuinely didn't know what to do, that he wasn't smart enough to understand that he needed to point to the other one. But it's also possible that it's a Homer Simpson-like effect, where when asked to point to what he wants, he just couldn't help but point to the one he wanted. And that the extra work it takes to lie was beyond him. And, in support of the second alternative, even adults find these tasks involving lying and deception more difficult. They were slower at them. We make more mistakes than tasks that don't involve lying and deception. So, I'm raising this not to solve the problem. You'll read more about it in the Peter Gray textbook and more about it in The Norton readings on development, but just to raise this as an interesting area of debate.

Another interesting area of debate is, "What's the relationship between different sorts of development?" So, I started off with Piaget, and Piaget, like Freud, believed in general, across the board changes in how children think. An alternative, though, is that there's separate modules, and this is a view developed, again, by Noam Chomsky, and also by the philosopher of mind Jerry Fodor, who claimed that the whole idea of a child developing as a single story is mistaken. What you get instead is there are separate pre-wired systems for reasoning about the world. These systems have some built-in knowledge, and they have to do some learning, but the learning pattern varies from system to system and there's a separateness to them. Why should we take this view seriously? Well, one reason is that there are developmental disorders that seem to involve damage to one system but not to another. And the classic case of this is a disorder known as autism. And autism is something I've always found a fascinating disorder for many reasons. It's actually why I entered psychology. I started off working with children with autism. And it could be taken as a striking illustration of how the social part of your brain is distinct from other parts of your brain.

So, what autism is is a disorder that strikes about one in a thousand people, mostly boys. And the dominant problems concern — consist of a lack of social connectedness, problems with language, problems dealing with people, and more generally, a problem of what the psychologist, Simon Baron-Cohen has described as "mind blindness." In that autistic people show no impairments dealing with the physical world, they show no impairments on — they don't necessarily show any impairments on mathematical skills or spatial skills, but they have a lot of problems with people. Now, many autistic children have no language; they're totally shut off from society. But even some of them who'd learned language and who managed to get some sort of independent life, nevertheless will suffer from a severe social impairment. And this could be shown in all sorts of ways.

A simple experiment developed by Simon Baron-Cohen goes like this. You show this to three- and four-year-olds. There's four candies there, and you say, "This is Charlie in the middle. Which chocolate will Charlie take?" For most children and most of you, I hope, the answer's pretty clear: This one.

Autistic children will often just shrug, say, "How could I know?" because they don't instinctively appreciate that people's interests and desires tend to be attuned to where they're looking.

Another sort of task, which is a task that's been done hundreds, perhaps thousands of times, is known as "the false-belief task" and here's the idea. You show the child the following situation. There's a doll named Maxie and Maxie puts the ball in the cupboard. Maxie leaves and a second doll enters. The second doll takes the ball out of the cupboard and puts it under the bed. Maxie comes back and the question is, "Where will Maxie look for the ball?" Now, this is a question about your understanding about minds. The question of where is the ball really is a question about the physical world. Everyone can solve it, but this question is hard. The right answer is Max will — Maxie will look in the cupboard, even though it's not really there because Maxie has a false belief about the world. Three-year-olds find this difficult. Two-year-olds find this difficult. Four-year-olds and five-year-olds are able to pass this task. Normal adults are able to pass this task. Children with autism have serious problems. And often, people with autism who are otherwise very high functioning will fail this task. They'll say, "Oh, he must think it's not — He'll — He's going to check under the bed."

Chapter 6. Question and Answer on Autism [00:40:40]

Any questions about autism? Yes.

Student: [inaudible]

Professor Paul Bloom: Good question. It isn't. They're both experiments designed to tap an appreciation of false belief. The deception one with the shoes and everything looked at it in the course of deception. Can you understand that the mother might think it's that person even though it's really that person? And our kid failed. This is a sort of stripped-down version without all the fanciness but it tests exactly the same thing. Yes.

Student: [inaudible]

Professor Paul Bloom: Nobody knows, but there's a theory which won't answer your question but will put it into a broader context. Simon Baron-Cohen argues that there are certain abilities that tend to be more sequestered for males, and other abilities that are more sequestered, more focused on females. Social abilities, he argues, tend to be more female than male. So, the way Baron puts it, provocatively, is to be a man is to suffer from a very mild form of autism [laughter]. The idea is then that autistic individuals suffer from what he calls extreme male brains, and as such, it stands to reason that they'd be more sampled from the male population than the female population. That's such an interesting issue, that again, when we return to talk about sex differences we'll look at that in a little bit more detail to see if it's supported by the evidence. Yes.

Student: [inaudible]

Professor Paul Bloom: I'm sorry. Tell me the — Is the severity of autism...

Student: [inaudible]

Professor Paul Bloom: It's an interesting question. The question is, "How do you think about the severity of autism with regard to developmental stages?" And sort of surprisingly, autism can't really be thought of in that way. So, it's not like an adult with autism is like a three-year-old or a two-year-old.

In some ways, somebody with autism isn't like any child at all, any normally developing child at all. So, it's not really a developmental delay in the way that it might make sense to think about certain forms of retardation. On the other hand, when we think about how severe autism is we do look at things like how much language does the person have, and in that sense, it is related to development. Yes.

Student: What are the chances that someone who's autistic would be able to overcome their deficiencies?

Professor Paul Bloom: The majority of people with autism. It's a good question. The question is, "What are the chances that somebody with autism will be able to overcome their deficiencies?" Autism is a funny disorder in that there's a lot of media publication and media presentation. Often the people who are showcased in the media tend to be very exceptional. So, there's a woman, Temple Grandin, who's autistic and — Has anybody here heard of Temple Grandin? She wrote some wonderful books about her experience as an autistic person, but she's very unusual. So a lot depends, to answer your question, how one defines autism, and whether one includes Asperger syndrome, which is a limited, a more mild syndrome, as a form of autism. The answer is that the majority of people with autism have severe problems, and will not, and at this stage, with this level of therapy, will not lead a normal life.

Student: More specifically, what I meant was, when you showed the example of Rain Man, ere they exceptional [inaudible]

Professor Paul Bloom: Right. The question is about so-called autistic savants. So, Rain Man, the character played by Dustin Hoffman, had extraordinary mathematical abilities. And some people with autism have extraordinary artistic abilities or mathematical abilities or musical abilities and these are amazing. It's an amazing question why they have it but this is a very small minority. This is a very — It's fascinating that it happens at all, that you have severe damage but compensated with some powerful skill. Now, I know I'm answering your question I think in a better way, but it's actually very rare. Most people with autism do not have any exceptional abilities that go along with it.

Another question is if you believe in modules — If there are modules, what are they? And so far when reviewing the developmental data we've talked about two of them: physics and people. An object module and a social module. But other people have argued that there is a special module in your brain for dealing with artifacts, that is, things like tables and chairs and cars and forks. Some people have argued there's a module for sociology, for dealing with human groups, races and classes and so on. Some have even argued that there is an intuitive biology, a common-sense biological understanding of the world that's separate from your understanding of people and physics. And, in fact, the most dominant proponent of the view is our very own Frank Keil, Master of Morse College at Yale, who has strongly defended the notion of an intuitive biological module.

Final question, just to raise: I've talked in terms of the modular view but there might also be profound general differences between children and adults, not just specific to how you think about objects or how you think about people or how you think about this or how you think about that, but rather more general. And one claim, which we're going to return to briefly next class when we talk about language, is that there's a very, very big difference between a creature that doesn't have language and a creature that does. And part of the claim is that learning a language, learning to speak, reconfigures the human

brain in such a way that is really exceptional. And that has no parallel in any other species. And this is an interesting claim and one we'll talk about.

Finally, I want to end with an example from Stephen Jay Gould. Suppose you hate development; you hate developmental psychology; you hate babies; you hate children; they're not cute; they're ugly; you don't want to have them; you don't want to study them; you're annoyed that we have to discuss them. Fine. But there are reasons to study development even if you are not interested in children because sometimes developmental studies and developmental data and developmental science can inform questions about adults.

And Stephen Jay Gould has a very nice example of this. He asked the question "Is a zebra a black animal with white stripes or a white animal with black stripes?" Now, you could look at adult zebras all day long and you're never going to figure this out. But if you want to know the answer, and I knew it, but I forget what it is — It doesn't matter. But if you wanted to know it you could. You would look at development and you'd watch the embryological development of a zebra and that's how you would learn the answer to your question. In fact, I'll end with a nice quote. This is by the famous biologist, D'Arcy Thompson, who wrote the book On Growth and Form, and it's sort of the model of many developmental psychologists and many evolutionary psychologists so I'll end with this: "Everything is the way it is because it got that way." Okay. I'll see you next week.